This document is provided solely as a complementary description of the proposed project. The information required for the official evaluation is contained exclusively in the CNPq submission system, in accordance with the call guidelines.

The purpose of this document is to provide international collaborators with a more detailed description of the project in English, facilitating their understanding of the proposed activities and expected outcomes. It also serves as supporting material for inviting additional international collaborators in the event that the proposal is approved.

CNPq/FNDCT Call No. 06/2026 – UNIVERSAL

Track A: Research groups led by researchers with a recent doctorate and without a statutory or CLT (labor-law) employment relationship with the institution executing the project

Annex I – Descriptive Outline of the Proposal

1. PROJECT

Title of the project:	**LAKE TWIN** - Development of digital twins for forecasting the hydrodynamics and water quality of tropical reservoirs
Proponent and coordinator:	Rafael de Carvalho Bueno
Participating institutions:	Federal University of Paraná
Graduate program of affiliation of the proponent:	Graduate Program in Environmental Engineering (PPGEA)
Lattes CV:	http://lattes.cnpq.br/7638962066048654
E-mail:	decarvalhobueno@gmail.com
Telephone:	+55 41 – 99905 3384
Keywords	Reservoirs, Environmental monitoring, Limnology, Hydrodynamic modeling
Predominant field of knowledge:	Environmental Sciences
Related fields of knowledge:	- Fluid Mechanics - Hydraulics

2. EXECUTING TEAM

Name	Institution	Degree	Role in the Project	Lattes CV or ORCiD
Rafael de Carvalho Bueno	UFPR	PhD	Coordinator and BCB Fellow	http://lattes.cnpq.br/7638962066048654

Graduate program and partner institutions:

Name	Institution	Degree	Role in the Project	Lattes CV or ORCiD
Tobias Bernward Bleninger	UFPR	PhD	Researcher (modeling)	http://lattes.cnpq.br/5131189316971564
Michael Mannich	UFPR	PhD	Researcher (sensors)	http://lattes.cnpq.br/0160328557166941
International collaborators				
Andreas Lorke	RPTU*	PhD	Researcher (physical limnology)	https://orcid.org/0000-0001-5533-1817

*RPTU = University of Kaiserslautern-Landau (The institution's name was recently changed; the former name, as also registered in the CNPq directory of institutions, was University of Koblenz-Landau).

3. INTRODUCTION AND OBJECTIVES

The growing pressure on water resources, associated with population growth, the expansion of economic activities, and the intensification of extreme climatic events, has increased the challenges related to the management of lakes and reservoirs (Sulistyowati et al., 2017). These systems play a strategic role in public water supply, energy generation, irrigation, recreation, and the maintenance of biodiversity, constituting central elements for water security and socioeconomic development. In Brazil, this dependence is particularly pronounced: a significant part of the metropolitan regions has its supply ensured by headwater reservoirs, whose operation continues to be considered strategic by the National Water Security Plan (PNSH).

Despite this relevance, reservoir management in Brazil remains, to a large extent, based on past information, involving spot monitoring, sporadic field campaigns, and models applied retrospectively to characterize an already-occurred state of the aquatic system. This reactive approach offers limited capacity to anticipate changes in thermal dynamics, circulation, and water quality, compromising operational planning in the face of extreme hydrological and climatic events, prolonged droughts, heat waves, and intense precipitation, whose frequency and intensity have changed in recent decades.

The general objective of this project is to develop a platform based on the concept of digital twins (*Digital Twins*) for the representation, forecasting, and intelligent management of the hydrodynamics and water quality of lakes and reservoirs (Figure 1). The project proposes to integrate continuous environmental monitoring, mathematical modeling, and machine learning techniques to create a dynamic virtual representation of these environments, capable of reproducing their current state, learning from observed data, and predicting their future evolution.

The digital twin will be fed by data obtained from an environmental monitoring network installed in the reservoir, including hydrodynamic, thermal, and water quality variables. These data will be continuously assimilated into models capable of representing fundamental processes of lake dynamics, such as the formation and evolution of thermal stratification, thermocline displacement, generation and propagation of internal waves, wind-induced circulation, vertical mixing, and transport of environmental properties.

The integration between hydrodynamic models and machine learning models will make it possible to develop tools capable of identifying patterns, reducing uncertainties, and generating forecasts of the evolution of the aquatic system. In this way, the project intends to advance from an approach based on the characterization of the current state of the reservoir to a predictive limnology approach, allowing the anticipation of changes in thermal dynamics and water quality and supporting water resource management strategies.

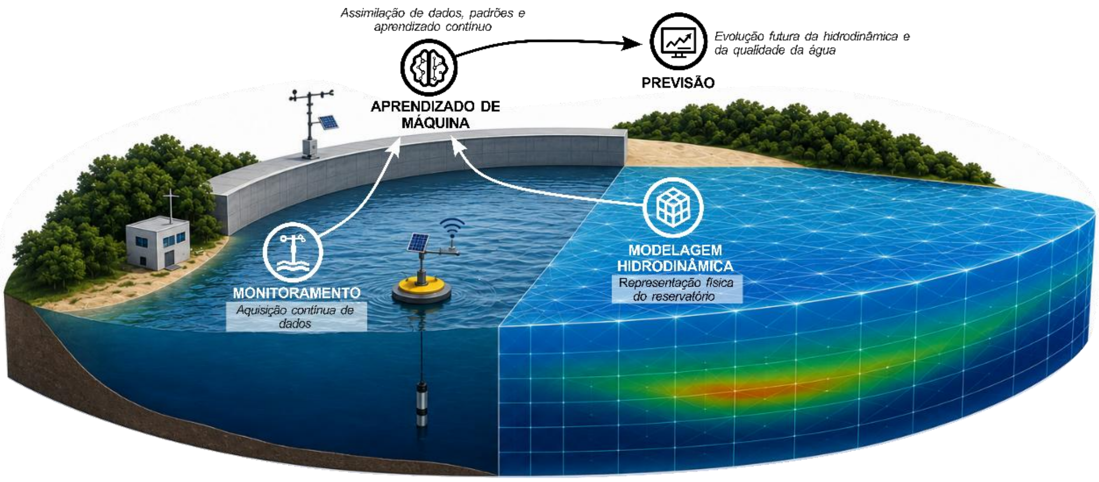

Figure 1. Illustrative diagram of the proposed monitoring system, intended to support the management of urban hydrology and the water quality of rivers and reservoirs.

In addition to the development of the reservoir's virtual environment, the project foresees the expansion of the Interwave Analyzer (de Carvalho Bueno, Bleninger, & Lorke, 2020), a tool for the analysis of lakes and reservoirs developed by the proponent of this proposal[1], incorporating new modules aimed at the integration between observational data, hydrodynamic simulations, and predictive algorithms. These tools will make it possible to improve the interpretation of the physical processes that govern water quality in tropical lakes and reservoirs.

Although the initial development is carried out in the Passaúna reservoir, the proposed methodology will be structured to allow its application in different lentic systems, contributing to a new approach to monitoring, forecasting, and sustainable management of lakes and reservoirs.

The specific objectives of the project are:

- To develop a unified high-temporal-resolution monitoring infrastructure for the continuous acquisition of environmental data in lakes and reservoirs, including parameters essential for the calibration, validation, and updating of the proposed models, such as water temperature, reservoir level, inflow and outflow discharges, and meteorological variables associated with the energy exchange processes between atmosphere and water, including solar radiation, wind speed and direction.
- To develop and validate hydrodynamic models of different scales for representing the dynamics of lakes and reservoirs, integrating morphometric characteristics, meteorological conditions, and observational data to simulate the thermal structure, internal circulation, and mixing processes.
- To develop a near-real-time data assimilation system capable of integrating observations from environmental monitoring into the hydrodynamic models, allowing the continuous updating of the reservoir state and the improvement of forecasts.

[1] https://buenorc.github.io/pages/interwave.html

- To develop a hybrid approach based on the integration between physical models and machine learning techniques for forecasting the temporal evolution of the dynamics and water quality in reservoirs.
- To evaluate the predictive capacity of the LAKE TWIN system to represent changes in thermal structure, water column stability, mixing processes, oxygen availability, and conditions favorable to the occurrence of events associated with water quality degradation, such as algal blooms
- To integrate into the LAKE TWIN system tools developed on the Interwave Analyzer platform, incorporating indicators of lake dynamics, including thermal stratification, thermocline depth, water column stability (e.g., Lake number and Schmidt stability), and characterization of internal waves, allowing an integrated interpretation of the physical processes and their effects on water quality.

4. PROJECT RELEVANCE AND JUSTIFICATION

In Brazil, this reality takes on even greater importance due to the strong dependence on reservoirs for public water supply and for regulating water availability. The National Water Security Plan (PNSH) identifies several strategic systems whose operation directly influences the supply of large urban centers and water resilience in the face of climate change. In the Metropolitan Region of Curitiba, for example, the supply of approximately four million inhabitants depends on a set of headwater reservoirs, among which Passaúna plays a fundamental role (Figure 2). The high dependence on these systems shows that improvements in monitoring and forecasting capacity do not represent merely scientific advances, but strategic tools to increase regional water security and reduce vulnerability to extreme events.

Although this proposal uses the Passaúna reservoir as a case study, its objective is not to develop a solution exclusive to this reservoir. Passaúna was chosen because it brings together a consolidated experimental infrastructure, historical data series, previously developed hydrodynamic models, and a research history that significantly reduces the scientific risks of the project. In this way, the reservoir functions as a natural laboratory for the development, validation, and demonstration of the proposed technology. The modular architecture of LAKE TWIN will allow the methodology to be subsequently adapted to other Brazilian reservoirs, especially those considered strategic for public water supply by the PNSH.

Tropical reservoirs present high hydrodynamic complexity due to the interaction between meteorological forcings, morphometric characteristics, thermal stratification, wind-induced circulation, vertical mixing, and biogeochemical processes (Bouffard et al., 2013; de Carvalho Bueno, Bleninger, et al., 2023; de Carvalho Bueno et al., 2026; B. E. Laval et al., 2005). These processes control the distribution of heat, nutrients, dissolved oxygen, and sediments, directly influencing water quality and the operation of supply systems. Changes in their dynamics can favor algal blooms, oxygen depletion, increased treatment costs, and reduced water availability (Flood et al., 2021; Pannard et al., 2011).

Despite recent advances in environmental monitoring and numerical modeling, reservoir management remains predominantly based on past information, offering limited capacity to anticipate future changes. At the same time, advances in environmental sensors, hydrodynamic models, data assimilation, and artificial intelligence have created the basis for the development of digital twins, virtual representations continuously updated by observational data capable of reproducing and predicting the behavior of physical systems (Yang et al., 2024) Although this technology has already been applied in different areas of engineering, its use for the operational forecasting of tropical lakes and reservoirs still remains incipient, representing an important scientific and technological opportunity.

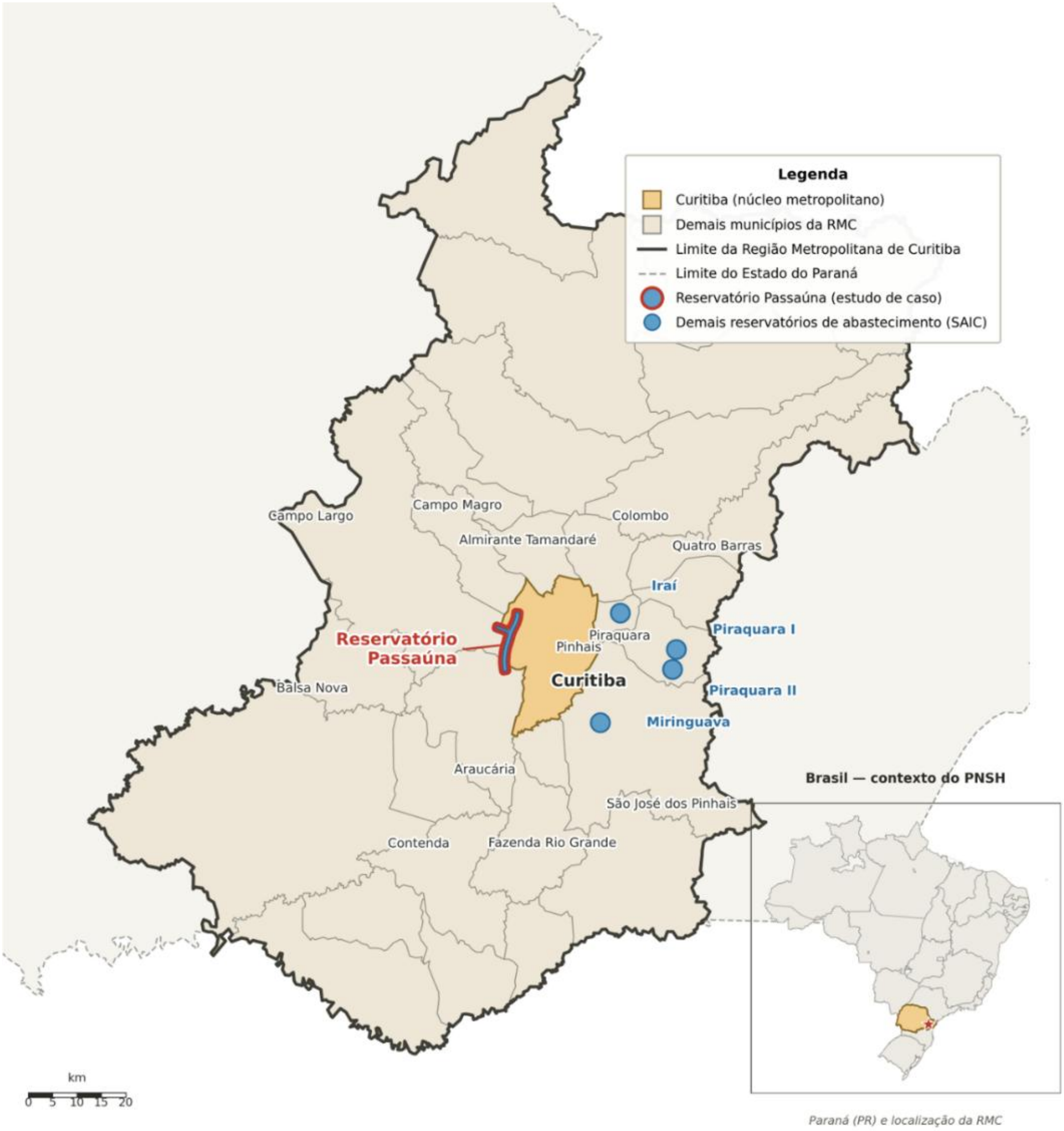

Figure 2. Location of the main water-supply reservoirs of the Metropolitan Region of Curitiba, highlighting the Passaúna reservoir, used as a case study in this proposal. In detail, the location of the region in the context of the National Water Security Plan (PNSH), highlighting the strategic importance of headwater reservoirs for regional water security.

The construction of digital twins for reservoirs represents a paradigm shift in the way these systems are monitored and managed. By integrating real-time observations, hydrodynamic models, and machine learning techniques, it becomes possible not only to reproduce the current state of the aquatic environment, but also to predict its future evolution, evaluate alternative scenarios, and identify environmental risks in advance. This predictive capacity can contribute significantly to the operational planning of reservoirs, the protection of water quality, adaptation to climate change, and the strengthening of water security.

In this context, the LAKE TWIN project proposes the development of a platform based on digital twins for tropical reservoirs, integrating environmental monitoring, hydrodynamic modeling, data assimilation, and machine learning. The platform will be developed as an interactive web application, using technologies for *front-end* development (HTML5, CSS, and JavaScript), combined with scientific visualization libraries and Web Geographic Information Systems (WebGIS), allowing the spatial and temporal representation of reservoir conditions. The system will provide, in near real time, interactive maps of the Passaúna reservoir containing observed and forecast variables, such as water temperature, thermal stratification, water column stability, circulation, level, water quality, and other hydrodynamic indicators, offering an intuitive interface for the continuous consultation of current conditions and the forecasts generated by the digital twin.

The specific scientific gaps and the expected advances will be detailed in the goals and objectives of the project.

5. GOALS AND DELIVERABLES

To achieve the proposed objectives, the project will be structured into four main goals, ranging from the acquisition and integration of environmental data to the development of an intelligent digital twin for forecasting the dynamics and water quality in tropical reservoirs:

- Goal 1: Implementation of the monitoring and environmental data acquisition infrastructure;
- Goal 2: Development of the hydrodynamic model and validation for operational forecasting;
- Goal 3: Development of data assimilation and machine learning systems for updating and forecasting the reservoir state;
- Goal 4: Platform LAKE TWIN for analysis, simulation, and forecasting of the dynamics and water quality.

Study area

The Passaúna reservoir (Figure 3), located in the Metropolitan Region of Curitiba, Paraná, was selected as a case study for the development and validation of the LAKE TWIN platform. The system constitutes one of the main sources of public water supply in the region, playing a strategic role for the water security of hundreds of thousands of inhabitants.

The Passaúna River Basin is located in the western portion of the Metropolitan Region of Curitiba, within the Upper Iguaçu basin. The Passaúna River is approximately 57 km long, with headwaters located in the Serras de São Luiz do Purunã and Bocaina, at about 1,040 m of altitude (da Silva Filho & Braga, 2009). In its middle-lower portion, in the late 1980s, a dam was built for public water supply, resulting in the formation of the Passaúna reservoir in 1990 (Rauen et al., 2018).

The reservoir has an approximate surface area of 9 km², a volume of about 6 × 10^7 m³, a length close to 11 km, and a maximum depth of approximately 17 m (Carneiro et al., 2016). Currently, it constitutes one of the main water supply sources of the Metropolitan Region of Curitiba, serving more than 650 thousand inhabitants.

Goal 1. Implementation of the monitoring and environmental data acquisition infrastructure

The development of a digital twin for reservoirs depends on the availability of environmental time series capable of representing the main physical processes that control the dynamics of the aquatic system. Although the Passaúna reservoir already has different monitoring and modeling initiatives developed in previous projects (Figure 3), with several related publications (de Carvalho Bueno, Bleninger, et al., 2023; Ishikawa, Bleninger, et al., 2021; Ishikawa, Gonzalez, et al., 2021; Prandini et al., 2025), much of the available data was obtained in specific campaigns or independent systems, with different acquisition frequencies, formats, and storage methods. This fragmentation limits the integration of data into continuous predictive systems and near-real-time monitoring applications (Lal et al., 2024).

In this way, this goal aims to implement an integrated infrastructure for the acquisition, organization, and transmission of environmental data, combining existing historical data, information from global models, and continuous observations obtained from reference sensors and low-cost sensors. This approach will allow the construction of a robust database for the calibration, validation, and updating of the hydrodynamic models (Goal 2) and the predictive algorithms based on machine learning (Goal 3).

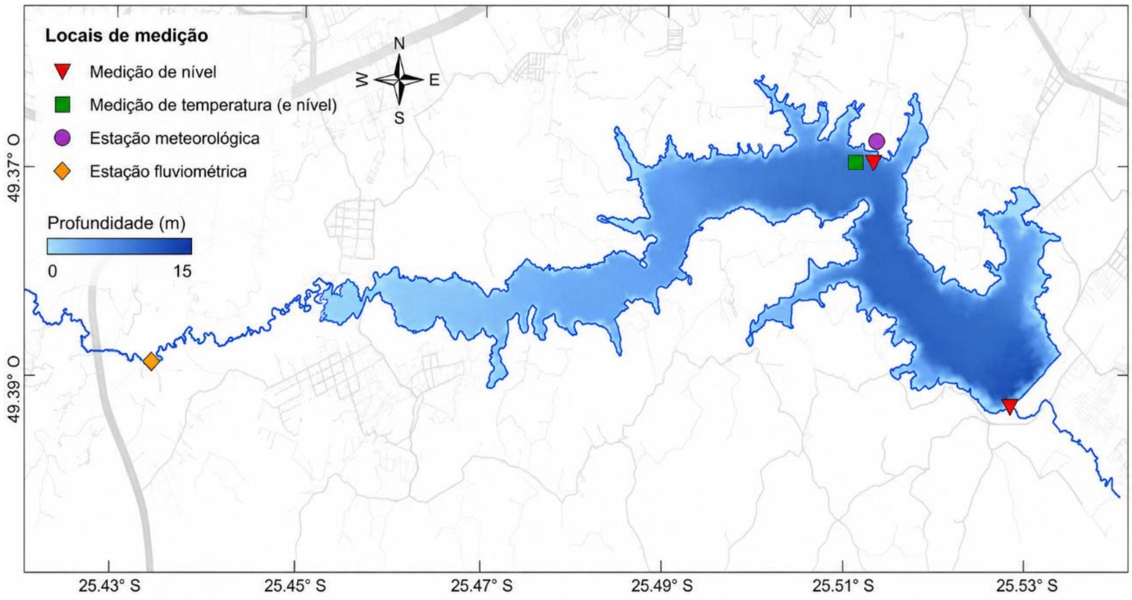

Figure 3. Bathymetric map of the Passaúna reservoir, indicating the current measurement sites.

Initially, an evaluation of the data available for the Passaúna reservoir will be carried out, including historical series of water temperature, reservoir level, inflow and outflow discharges, meteorological characteristics, and other variables necessary to represent the energy exchange, transport, and mixing processes in the aquatic environment. When necessary, this information will be complemented by data from meteorological reanalysis products, global hydrological models, and open-access environmental databases, allowing the generation of continuous series even in periods without direct observations.

In parallel, the integration of the sensors currently used in the reservoir monitoring will be evaluated, considering aspects such as acquisition frequency, data format, communication availability, and compatibility with an automated storage and transmission infrastructure. In cases where existing equipment has integration capacity, automatic communication with the proposed platform will be implemented. For systems without connectivity, acquisition and telemetry modules capable of performing data collection, temporary storage, and transmission may be developed.

In addition to the integration of the already available information, additional continuous monitoring sensors will be installed for parameters essential to updating the system, prioritizing the variables of greatest relevance to the reservoir's hydrodynamics, such as water level, water column temperature, and discharges associated with the main flows of the system. Depending on the availability of infrastructure resources and complementary funding, new sensors may be incorporated to expand the characterization of water quality and biogeochemical processes.

The additional sensors will be evaluated in the laboratory using reference equipment, in order to determine their precision, stability, and applicability in real environmental conditions. Data transmission will be based on low-energy-consumption technologies, such as LoRa, allowing greater flexibility for expanding the observational network in different regions of the reservoir.

The data obtained will be organized in an integrated database containing environmental time series, which will serve as support for the subsequent stages of the project, including hydrodynamic modeling, data assimilation, and development of the predictive system of the digital twin.

Goal 1:	*The main deliverable of this goal will be an integrated environmental database of the Passaúna reservoir, combining historical series, global model data, and continuous measurements obtained by connected sensors. In addition, an infrastructure for data acquisition and transmission capable of feeding near-real-time modeling and forecasting systems will be developed, together with protocols for the evaluation of low-cost sensors for environmental monitoring.*

Goal 2. Development of the hydrodynamic model and validation for operational forecasting

The construction of a digital twin for reservoirs requires, in addition to the availability of continuous environmental data (Goal 1), a robust mathematical representation capable of reproducing the fundamental physical processes responsible for the temporal and spatial evolution of the aquatic system. In this way, this goal aims to develop and validate a hydrodynamic representation of the Passaúna reservoir capable of simulating the thermal dynamics, internal circulation, and processes associated with water quality, constituting the physical basis for the proposed predictive system.

Reservoirs and lakes are complex systems in which processes occur at different spatial and temporal scales. Seasonal changes in thermal structure and energy balance control the formation and persistence of stratification (Mortimer, 1952), while short-duration events associated with wind, upwelling (*upwelling*) and lateral circulation can rapidly modify the distribution of physical and biogeochemical properties in the water column (de Carvalho Bueno, Bleninger, et al., 2023; de Carvalho Bueno et al., 2026; Lorke et al., 2002; Preusse et al., 2010). Therefore, the adequate representation of these systems requires a balance between physical complexity, spatial resolution, and computational efficiency.

In this goal, a hierarchical modeling approach will be developed, combining hydrodynamic models of different scales to meet the needs of operational forecasting and detailed analysis of the reservoir's physical processes. Initially, a vertical one-dimensional model will be implemented using the General Lake Model (GLM) to continuously represent the evolution of the reservoir's thermal structure, including stratification, water column stability, thermocline depth, and vertical mixing processes.

The use of a one-dimensional model will allow high-frequency simulations at low computational cost, being suitable for continuous operation in the context of the digital twin with continuous data forecasting (Lin et al., 2022). The model will be run in operational mode using sequential updates of the initial conditions, allowing the forecast of each period to be continuously updated from the previous state of the reservoir and the new meteorological and environmental conditions, a process similar to that carried out in Lake Erie (Lin et al., 2022).

Additionally, a three-dimensional hydrodynamic model will be developed using Delft3D FLOW, allowing the representation of spatial processes that are not adequately described by one-dimensional models.

The simulations will be carried out using high-resolution meteorological and hydrological forcings, with assimilation of the observational data from the infrastructure developed in Goal 1, in addition to data from operational meteorological and hydrological databases for generating future forecasts. Atmospheric forecast products will be evaluated, such as the Global Forecast System (GFS), the Integrated Forecasting System (IFS/ECMWF), the Global Deterministic Prediction System (GDPS), and regional meteorological forecasts, including variables such as wind speed and direction, solar radiation, air temperature, humidity, and precipitation. For future hydrological conditions, discharge forecast products may be used, such as the Global Flood Awareness System (GloFAS), or rainfall-runoff models applied to the reservoir basin, allowing the estimation of the inflow discharges necessary for the predictive operation of the hydrodynamic models. The combination of these sources will allow the continuous updating of the reservoir state and the generation of forecasts of thermal dynamics, circulation, and water quality.

Initially, the models will be calibrated using historical data from previous studies carried out in the Passaúna reservoir, including limnological campaigns and available time series (de Carvalho Bueno, Bleninger, et al., 2023; Ishikawa, Bleninger, et al., 2021; Ishikawa, Gonzalez, et al., 2021; Prandini et al., 2025). Subsequently, validation will be carried out using the continuous data obtained by the integrated monitoring network developed in this project.

The performance of the models will be evaluated using statistical metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and coefficient of determination, considering variables such as water temperature, thermocline depth, and water column stability. This stage will allow quantifying the models' capacity to reproduce the current state of the reservoir and identifying the main sources of uncertainty.

The integration of models of different scales will be evaluated and will allow establishing the most suitable physical architecture for the implementation of the LAKE TWIN system.

Goal 2:	*The main deliverable of this goal will be a validated hydrodynamic modeling architecture for the Passaúna reservoir, composed of models of different spatial and temporal scales, evaluated regarding their capacity to reproduce the thermal and hydrodynamic dynamics of the system for application in near-real-time monitoring and forecasting. In addition, a scientific article will be developed comparing the performance of the one-dimensional and three-dimensional models and evaluating the need for integration between different levels of complexity for the development of digital twins applied to tropical reservoirs.*

Goal 3. Development of data assimilation and machine learning systems for updating and forecasting the reservoir state

One of the main challenges for the implementation of digital twins applied to environmental systems is to ensure that the virtual representation remains continuously synchronized with the real system (Yang et al., 2024). Hydrodynamic models are capable of representing the fundamental physical processes of lakes and reservoirs (Hodges et al., 2000; B. Laval et al., 2003), but they present uncertainties associated with mathematical simplifications, initial conditions, and limitations of the input data. In this way, this goal aims to develop an integrated data assimilation and machine learning system capable of continuously updating the reservoir state and improving its forecasting capacity (Baracchini et al., 2020).

The proposed approach will be based on the integration between the hydrodynamic models developed in Goal 2, the continuous data from the monitoring infrastructure of Goal 1, and artificial intelligence techniques. The system will be structured in a digital twin architecture composed of an environmental data layer and a model layer, allowing real observations to be used to correct and update the state of the virtual model in continuous cycles.

Initially, a data assimilation system capable of integrating the observations obtained by the sensors into the hydrodynamic model will be implemented. Assimilation techniques based on statistical filters (Baracchini et al., 2020), such as the Ensemble Kalman Filter (EnKF), will be evaluated for updating the main state variables of the reservoir, including water temperature, thermal structure of the water column, reservoir level, and parameters related to water quality. This process will reduce discrepancies between the simulated and observed system, keeping the digital twin continuously updated.

Data assimilation will allow establishing an operational cycle in which the hydrodynamic model initially generates a forecast of the future state of the reservoir using available meteorological and hydrological conditions. Subsequently, the new observations obtained by the sensors will be used to correct the forecast state, generating an updated condition that will serve as the basis for the subsequent forecast cycles.

In addition to assimilation based on statistical methods, machine learning models capable of identifying temporal patterns and non-linear relationships between the monitored environmental variables will be developed. Deep learning techniques for time series will be evaluated for forecasting variables of interest related to the dynamics and water quality, such as temperature evolution, water column stability, and conditions favorable to the occurrence of critical events.

A hybrid approach based on physical knowledge and machine learning will be used, in which the results of the hydrodynamic models will be employed as a source of information for training the algorithms, reducing the need for large volumes of observational data and ensuring greater physical coherence in the forecasts. This strategy will make it possible to combine the capacity of mathematical models to represent fundamental processes with the capacity of artificial intelligence algorithms to capture complex patterns present in environmental data.

The system will be developed considering data from multiple sources, including in situ measurements, meteorological data, atmospheric forecasts, and remote sensing products. This integration will make it possible to expand the predictive capacity of the digital twin, reducing limitations associated with the restricted spatial coverage of the point sensors installed in the reservoir.

The developed models will be evaluated regarding their forecasting capacity using different time horizons, comparing the results predicted by the system with independent validation data. Forecast errors, operational stability, and the capacity to anticipate changes in the thermal dynamics and water quality of the reservoir will be evaluated.

Goal 3:	*The deliverable resulting from this goal will be the SMART platform, capable of continuously and remotely monitoring the water quality and dynamics of the Passaúna reservoir. The platform will integrate data from low-cost sensors and hydrodynamic models, transmitting the information to a web server accessible both to the public and to the authorities. In addition, a detailed manual will be prepared for the reimplementation of the system in other regions, with the possibility of future expansions to incorporate new sensors and additional models.*

Goal 4. Platform LAKE TWIN for analysis, simulation, and forecasting of the dynamics and water quality

The final stage of the project will consist of the integration of the hydrodynamic models, data assimilation systems, and predictive algorithms developed in the previous goals into an operational digital platform for the Passaúna reservoir. The LAKE TWIN platform will be developed as an online interface capable of continuously representing the current and future state of the reservoir, allowing the integrated visualization of observational data, hydrodynamic simulation results, and forecasts obtained by the digital twin (Figure 4).

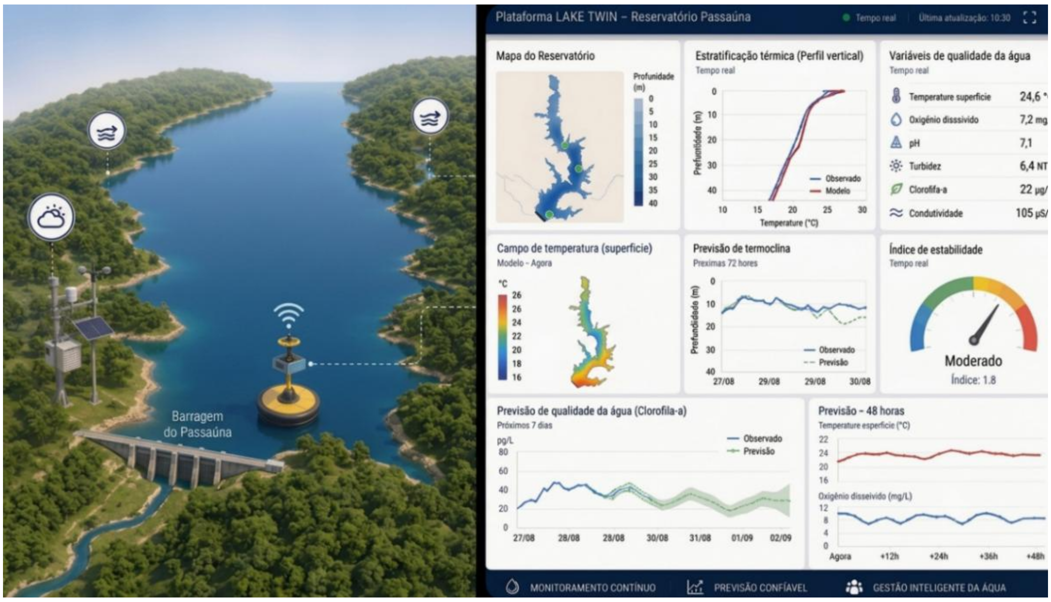

Figure 4. Illustrative diagram of the LAKE TWIN forecasting platform, which will integrate the sensor-based monitoring system with the forecasting system obtained through the mathematical models.

The platform will allow continuous monitoring of the main hydrodynamic and water quality variables, including the spatial and temporal distribution of temperature, thermal structure of the water column, stratification, circulation, mixing processes, reservoir level, and indicators associated with water quality. The results will be presented through maps, vertical profiles, time series, and products derived from the models, allowing the tracking of the reservoir's evolution and the identification of changes in its environmental state.

In addition to the variables directly simulated by the hydrodynamic models, the platform will incorporate analytical tools developed on the Interwave Analyzer platform (de Carvalho Bueno, Bleninger, & Lorke, 2020), allowing the automatic calculation of limnological and hydrodynamic indicators, such as thermocline depth, Schmidt stability, Lake number, potential energy available for mixing, Brunt-Väisälä frequency, natural oscillation periods, internal wave modes, and other parameters associated with the stability and internal dynamics of lakes and reservoirs. These indicators will be calculated both for the observed data and for the scenarios predicted by the digital twin, allowing the early evaluation of changes in water column stability, mixing events, intensification or weakening of thermal stratification, and other physical processes that influence water quality.

Computational architecture will be developed in a modular way, allowing the incorporation of new analysis parameters, mathematical models, predictive algorithms, and analytical tools. This approach will allow the future expansion of the platform to other lakes and reservoirs, constituting a broad infrastructure for the monitoring, analysis, and forecasting of hydrodynamics and water quality based on the concept of digital twins.

Goal 3:	*The main deliverable of this goal will be the online LAKE TWIN platform, integrating observational data, hydrodynamic models, and predictive systems into a digital environment for the monitoring, simulation, and forecasting of the dynamics and water quality of reservoirs. In addition, a conference scientific article will be developed presenting the architecture and application of the digital twin to support the management of tropical reservoirs.*

6. ABOUT THE PROPOSAL COORDINATOR AND PREVIOUS STUDIES

The coordinator of this proposal, Rafael de Carvalho Bueno, holds a doctorate in Environmental Engineering from the Federal University of Paraná with a double degree in Natural Sciences from the University of Kaiserslautern-Landau, Germany. Rafael de Carvalho Bueno has experience in physical limnology, with topics involving the dynamics of lakes and reservoirs (de Carvalho Bueno, Bleninger, & Lorke, 2020; de Carvalho Bueno, Bleninger, et al., 2023, 2024b, 2024a; de Carvalho Bueno, Bleninger, Yao, et al., 2020; de Carvalho Bueno, da Silva, et al., 2024; de Carvalho Bueno et al., 2026; de Carvalho Bueno, Ríha, et al., 2023). Rafael was also responsible for developing the *Interwave Analyzer* (de Carvalho Bueno, Bleninger, & Lorke, 2020), a tool intended for the analysis of the hydrodynamics of lakes and reservoirs based on the thermal and meteorological characterization of the environment under analysis, with several examples of applications in lakes around the world (Bouffard et al., 2025; Chesnokov et al., 2025; de Carvalho Bueno, Ríha, et al., 2023; Guseva et al., 2021; Koue, 2025; Prandini et al., 2025; Shi et al., 2023, 2025; Tsydenov, 2022). Together with other team members, he participated in several national and international projects carried out at the Passaúna Reservoir, through an agreement between the Federal University of Paraná and SANEPAR - Paraná Sanitation Company (Cooperation Agreement 171/2019, process 23075.069175/2018-09). Among the most relevant projects are the MuDak-WRM, FotoÁgua, and LIGAS projects (Table 1).

In 2026, Rafael coordinated his first research project approved in a funding call as coordinator. The MIXLAKE project – Physical Limnology of Lakes and Reservoirs, awarded in CNPq Call No. 49/2024, investigated, through three-dimensional numerical hydrodynamic modeling, the influence of bathymetry on the dynamics of lakes and reservoirs and on the predominance of internal waves. Among the main results, the proposition of a critical topographic number to characterize the behavior of internal seiches in different reservoir geometries stands out, contributing to the understanding of the mixing, circulation, and turbulence processes in these ecosystems (de Carvalho Bueno et al., 2026). The project also promoted advances in the development of the Interwave Analyzer platform, incorporating new tools for the analysis of thermal stratification, water column stability, internal wave modes, and other hydrodynamic indicators, expanding its application in physical limnology studies.

The experience accumulated in the projects developed at the Passaúna reservoir allowed the construction of a solid scientific and technological basis, which culminates in this proposal. Table 1 summarizes the main projects conducted in the region, highlighting their objectives, contributions, and the way each one complements the previous ones. It can be observed that this project represents a natural evolution of these studies, integrating the knowledge generated in environmental monitoring, hydrodynamic modeling, and computational tools into a predictive platform based on the concept of digital twins.

Project Name:	Mudak	FotoÁgua	Probral	LIGAS	MixLake	LAKE TWIN
Call No.:	German Ministry of Research and Development, BMBF GROW Call	Call 17/2021 – Paraná State Program of Research in Environmental Sanitation PPPSA - 89/2022 PDI	CAPES-Probral 20222176949P	CNPq	CNPq 49/2024 - 152147/2025-0	CNPq/FNDCT No. 06/2026 - UNIVERSAL
Funding:	BMBF	PPPSA CNPq	CAPES	CNPq	CNPq	CNPq
Amount:	R$ 13.151.700,00	R$ 381.000,00	R$ 1.250.000,00	R$ 247.025,00	R$ 68.160,00	R$ 239.428,00
Coordinator:	Tobias Bleninger	Tobias Bleninger	Tobias Bleninger	Michael Mannich	Rafael de Carvalho	Rafael de Carvalho
Term:	2017 to 2020	2022 to 2023	2023 to 2024	2024 to 2026	2025 to 2026	2027 to 2028
Objectives:	The main objective was the investigation of limnological parameters and the development of a model for forecasting medium- and long-term changes in the water quality of reservoirs, one that would be as simple as possible and widely accessible.	The main objective was to analyze the effects of floating photovoltaic systems installed in reservoirs on air-water interface transfer processes and changes to limnological aspects.	The main objective was to evaluate the effects of floating photovoltaic systems in reservoirs, focusing on the reduction of evaporation, algae growth, and electricity generation, as well as to investigate their impacts on the water quality and ecology of reservoirs.	The general objective of the project was to investigate the effects of floating photovoltaic systems installed in reservoirs on air-water interface transfer processes and changes in limnological aspects through the monitoring of physical variables and gas fluxes at the surface.	The general objective of the project was to evaluate how the bathymetry of lakes and reservoirs controls the dynamics of internal seiches and mixing processes, using non-hydrostatic three-dimensional numerical simulations to investigate the generation, propagation, and dissipation of energy in different morphological scenarios.	The LAKE TWIN project aims to develop an intelligent digital twin for reservoirs, integrating continuous monitoring, hydrodynamic modeling, and machine learning for forecasting the dynamics and water quality.

Table 1. Summary of the information and objectives of each project already carried out in the Passaúna reservoir and tributaries.

The Passaúna reservoir was chosen as a case study due to the existence of a consolidated infrastructure for environmental monitoring, including continuous measurements of hydrodynamic and water quality parameters, in addition to a history of studies using reference sensors and field campaigns in the region. The availability of previously collected data and accumulated knowledge about the thermal dynamics, circulation, and water quality of the reservoir and its tributaries provides an adequate scientific and experimental basis for the development, calibration, and, mainly, validation of the proposed system.

Although several projects have contributed to the environmental and hydrodynamic characterization of the Passaúna reservoir (Table 1), none of them aimed to integrate continuous monitoring, computational modeling, and artificial intelligence into a single predictive structure capable of representing the current state and estimating the future evolution of the system. In this way, the LAKE TWIN project differentiates itself by proposing the construction of a digital twin of the reservoir, combining real-time observational data, hydrodynamic models, and machine learning techniques for forecasting the dynamics and water quality.

In addition, the developed monitoring network may constitute a permanent infrastructure for future studies in the region, allowing the continuity and integration of environmental data.

7. TECHNICAL SUPPORT AND PARTNERSHIPS

7.1. Technical support

The previous projects carried out in the Passaúna region play a fundamental role in enhancing this proposal, since many advances can be made by taking advantage of the already available infrastructure, mainly with respect to the use of reference sensors and monitoring platforms. Variables already monitored in previous projects (e.g., temperature, level, velocity, discharge, etc.) may be used by the mathematical models, allowing a broader measurement horizon and variability, mainly for the machine learning models. In addition, the existing infrastructure, technical support (such as vessels, engines, and the structure of the monitoring stations), and some reference sensors previously acquired will be available for use in this project, facilitating its execution and expanding the available resources.

The already available infrastructure that may contribute to the execution of this project includes:

- *Meteorological Station:* A high-precision station capable of monitoring parameters such as wind speed and direction, air temperature, relative humidity, incident solar radiation, and precipitation. Even though it cannot be automatically adapted to the telemetric system, it may be used to calibrate the station used in the project.
- *Fluviometric Station:* A fluviometric station located upstream of the reservoir. A station for monitoring the level in the spillway region. A station located in the intake region for monitoring the intake discharge and water level.
- *Available Laboratories:* Laboratories of the Graduate Programs in Environmental Engineering (PPGEA) and Water Resources and Environmental Engineering (PPGERHA) of the Federal University of Paraná, with access to multi-user equipment.
- *Vehicles:* Two Kombi vans and two 4x4 pickup trucks in excellent condition, which support the transport of teams in field activities.
- *Boat:* Three vessels with 15HP and 25HP engines, in addition to an individual kayak for short trips. There are also life jackets, paddles, and appropriate clothing for river activities.

- *Measurement Equipment:* Two multiparameter probes for monitoring physical parameters of water quality, such as pH, temperature, turbidity, and conductivity. The project also has a CTD to measure conductivity and temperature at different depths.
- *Basic Equipment:* Tools, anchors, ropes, batteries, transformers, inverters, lamps, life jackets, and two metallic floating platforms for installing sensors in the water.

7.2. International partnerships

This proposal will count on the collaboration of researchers from an international institution, with the objective of strengthening the internationalization of the research activities, expanding the scientific insertion of the project, and establishing a basis for future cooperation in national and international projects. The partnership will contribute to the exchange of knowledge, methodological development, and improvement of the approaches for monitoring, modeling, and forecasting the dynamics of lakes and reservoirs.

The collaboration will be established with the Physical Limnology research group of the University of Kaiserslautern-Landau, led by Andreas Lorke. The researcher has extensive international experience in the field of physical limnology, with relevant contributions related to lake dynamics, mass transport, mixing processes, and the interaction between physical processes and water quality.

The partnership has a consolidated scientific relationship with the proponent of this proposal, built during the sandwich doctorate period and maintained through previous research projects. Dr. Andreas Lorke participated in the scientific training of the proponent, contributing to the development of methodologies applied to the analysis of internal wave dynamics in lakes and reservoirs, including the development of the open-source Interwave Analyzer platform (de Carvalho Bueno, Bleninger, & Lorke, 2020), aimed at the characterization of internal oscillations and hydrodynamic processes in lake environments.

In the context of the LAKE TWIN project, the international collaboration will be fundamental for the improvement of the methodologies for hydrodynamic modeling, data assimilation, and integration between field observations and predictive models.

In addition to the expected scientific results, the partnership will make it possible to consolidate an international research network focused on the intelligent monitoring and forecasting of the dynamics of aquatic ecosystems, creating opportunities for the joint submission of future projects and the expansion of the developed system to other lakes and reservoirs.

8. EXPECTED RESULTS AND CONTRIBUTIONS

The LAKE TWIN project proposes the development of a digital twin for reservoirs, integrating continuous environmental monitoring, hydrodynamic modeling, and advanced data assimilation and machine learning techniques. The platform will make it possible to continuously represent the current state of the reservoir, update its conditions based on observational data, and generate forecasts of the evolution of the water dynamics and parameters associated with environmental quality. This approach represents an advance over traditional monitoring systems, which usually present limitations related to the low frequency of data acquisition and the absence of integrated tools for forecasting.

The main technological contribution of the project will be the creation of an integrated architecture capable of connecting environmental sensors, mathematical models, and intelligent algorithms into a single operational system. Unlike platforms aimed only at data visualization, LAKE TWIN will be capable of using near-real-time information to correct the state of the models and improve future forecasts, allowing applications aimed at monitoring thermal stratification, water column stability, internal circulation, and indicators related to water quality.

From a scientific point of view, the project will contribute to the advancement of applied physical limnology by integrating different scales of representation of reservoir dynamics. The combined use of hydrodynamic models of different complexities will make it possible to evaluate the balance between physical accuracy and computational efficiency, establishing a methodology for the operational forecasting of lake environments. In addition, the continuous assimilation of field data will make it possible to reduce uncertainties associated with the numerical models and improve the predictive capacity of the systems used in water resource management.

The developed platform will have the Passaúna reservoir as a case study, but its architecture will be designed to allow replication in other lakes and reservoirs subject to different environmental pressures. The integration between low-cost sensors, remote communication systems, and predictive models will make it possible to expand the monitoring of aquatic environments, especially in regions where traditional networks present limitations in spatial and temporal coverage.

In addition to the scientific and technological contributions, the project may strengthen the integration between research institutions, management agencies, and companies responsible for the operation of water systems. The development of a tool capable of providing predictive information about the dynamics and water quality may support decision-making processes related to water security, control of critical events, operational planning, and conservation of aquatic ecosystems.

The project will also contribute to the training of human resources specialized in strategic areas, including environmental instrumentation, data science, computational modeling, and artificial intelligence applied to water resources. The students involved will have contact with all stages of the development of an integrated technological solution, from the acquisition of data in the field to the implementation of computational systems for environmental forecasting.

In general, LAKE TWIN seeks to transform reservoir monitoring into a predictive approach, combining advances in sensing, mathematical modeling, and artificial intelligence to support a more efficient, sustainable, and adaptive management of water resources in the face of the challenges posed by climate change and the increase in anthropogenic pressures.

9. PROJECT EXECUTION SCHEDULE

The project schedule, presented below, covers all 4 years of the project, from the development of sensors and modeling to the implementation of the SMART platform, where the last year of the project will be dedicated to the final tests and improvement, not only of the SMART platform, but also of all the developed sensors, focusing on the evaluation of the maintenance and replacement times of each component of the system. The schedule is divided according to each goal of the project.

Goals	Activity description	Year 1						Year 2					
		1	2	3	4	5	6	1	2	3	4	5	6
1,2,3,4,5	Reports (partial and final)						■						■
1,2,3,4,5	Equipment acquisition	■	■	■									
1	Survey of measurement data carried out in past projects	■											
1	Analysis of existing sensors and specifications for monitoring	■	■										
1	Development of the telemetric system and additional sensors		■	■									
1	Calibration and validation of sensors with reference sensors		■	■									
1	Data acquisition and formulation of global and regional forecast models				■	■	■	■	■	■	■	■	■
1	Implementation and monitoring of the parameters				■	■	■	■	■	■	■	■	■
2	One-dimensional modeling for past data	■	■	■									
2	Three-dimensional modeling for past data	■	■	■									
2	Validation of models for current data and extrapolations with forecast models				■	■						$\mathcal{M}$	
2	Incorporation into the LAKE TWIN system						■	■					
3	Development of the machine learning model				■	■						$\mathcal{E}$	
3	Validation of the proposed models					■	■						
3	Incorporation into the LAKE TWIN system						■	■					
4	Development of the backend of the LAKE TWIN platform							■	■	■	■	$\mathcal{E}$	
4	Development of the frontend of the LAKE TWIN platform									■	■		
4	Continuity and improvement tests of the LAKE TWIN platform										■	■	■

Legend	Description
■	Planned period for the execution of the activity
■	Mandatory reports (required by CNPq)
$\mathcal{M}$	Preparation and submission of a scientific article to a journal
$\mathcal{E}$	Preparation and submission of a scientific article to an event

10. BUDGET

Below, all the expenses foreseen for this proposal are detailed, as well as the justifications for each budgeted item.

10.1. Per diems and travel

Item	Description	Per diem/ Travel (R$)	Duration (days)	Total Price (R$)
National conference (per diems)	2 participations in SBRH or similar event	320,00	(6/event)	7.680,00
		2.000,00	2 x Round trip	4.000,00
Field tasks	Work at Passaúna and tributaries for installation and maintenance of the system	320,00	12 (1/month for 1 year of project)	3.840,00
Total estimated value:			**R$ 15.508,00**	

SBRH = Brazilian Symposium on Water Resources

10.2. Operating costs: Consumables

Item	Description	Price (R$)	Total (R$)	Justification
Basic electronic components	General electronic components, such as microcontrollers, capacitors, resistors, transistors, diodes, LEDs, communication modules, various sensors, displays, relays, voltage regulators, solar panels and conversion modules, connectors, power supplies, batteries, solder tin, motors, among others	-	15.000,00	Essential for the assembly of the sensors and control systems. These components include microcontrollers and other elements fundamental to the communication and operation of the LAKE TWIN platform.
General components for field work	General components, such as ropes, sensor supports, screws, steel rods and plates, buoys, etc.	-	10.000,00	Items such as tools, supports, and devices necessary for the installation and maintenance of the sensors
Data storage system and online management of the SMART platform	2-year plan (24 months) for Azure Cosmos DB or similar and OpenAI API or similar	850,00 /month	20.400,00	Fundamental for the management, processing, and storage of the collected data and maintenance of the LAKE TWIN platform

Item	Description	Price (R$)	Total (R$)	Justification
Total estimated value:			**R$ 45.400,00**	

10.3. Capital: permanent material

Item	Description	Price (R$)	Total (R$)	Justification
Desktop Computers	1 Desktop computer DELL XPS 14th generation Intel® Core™ i7-14700 with a Dell 27" Monitor or similar	15.000,00 /unit	15.000,00	Used for data processing and carrying out the work by the students and researchers involved in the project
Total estimated value:			**R$ 15.000,00**	

10.4. Scholarships

Modality	Duration (months)	Qty.	Unit value (R$)	Total Value (R$)
PDJ Scholarship	24	1	-	136.320,00
IC Scholarship	24	1	700,00	16.800,00
Total estimated value:			**R$ 153.120,00**	

*Only the IC scholarship must be counted toward the R$ 100,000.00 limit

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